

July 1, 2026

Dear Associate Professor Olivier Haas,

I am writing to express my strong interest in pursuing a PhD in AI for automotive systems at Coventry University. With its focus on designing verifiable, safe and interpretable multimodal large language model control system, I believe this opportunity aligns perfectly with my research interests and skills.

Having recently graduated as an AI & Data Science Engineer, I have gained significant experience in developing innovative solutions that leverage machine learning and data analysis techniques. My most recent project, Cardiovascular Medical Digital Twin, showcases my ability to design robust and efficient models for real-time inference. Specifically, my implementation of the linear-time state space model, Medical_MambaTS, demonstrates my expertise in creating low-latency systems that can handle complex interactions between multiple sensors and control systems.

This experience resonates with the PhD offer's requirement for designing a predictive control framework that accommodates interaction between multiple control systems exploiting vehicle environment understanding. I am excited about the prospect of applying my skills to develop a novel multimodal large language model control system that ensures safety, interpretability, and verifiability in autonomous vehicles.

Furthermore, my work on SST_KODA_MultiScale has shown me the importance of robust cloud forecasting in handling missing data and sensor dropouts. I believe this expertise will be valuable in tackling the challenges associated with designing a control system that can dynamically handle missing data and sensor dropouts.

I am confident that my unique blend of engineering background, research experience, and passion for AI and data science make me an ideal candidate for this PhD program. I would welcome the opportunity to discuss my application and how my skills align with the research objectives.

Thank you for considering my application. Please find attached my CV, academic references, and the one-page summaries of both my Engineering and

Master's theses. I look forward to hearing from you soon.

Sincerely,

Ismail Aissaoui